A Sharp Schur-Positivity Transition for Four-Port Clique Attachments

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Abstract

We classify Schur positivity for a five-parameter family of quasi-line graphs obtained by attaching a clique to four independent vertices of a fixed twelve-vertex line graph. The four attachment groups have arbitrary positive integer sizes; a fifth group has no neighbors in the original graph. Schur positivity is equivalent to the nonnegativity of an explicit quartic, and failure of positivity produces exactly one negative Schur coefficient. The classification is computer-assisted: a finite transfer on the original graph and a four-corner reduction produce exact polynomial sign certificates covering all parameters. We also give an analytic description of the dilution transition. Its sharp normalized threshold is $(\sqrt{2}-1)/2$, approached precisely when one attachment pair becomes negligible and the other becomes balanced. An exact finite certificate locates the smallest integral transition at graph order 48.

Contribution and significance

The underlying twelve-vertex counterexample and the existence of infinite non-Schur-positive families are prior results. This paper instead classifies the complete positive and negative regions for a specified four-port attachment family with five arbitrary integer parameters. One explicit coefficient controls every sign: its quartic is nonnegative exactly on the Schur-positive region. The other coefficients are covered by a finite exact certificate whose reduction to finitely many cases is proved, rather than inferred from a bounded search.

The classification also permits an analytic study of how unattached clique vertices destroy positivity. We determine the sharp normalized threshold, characterize all asymptotically extremizing allocations, and certify the smallest integral transition. Stable partitions, inclusion–exclusion, long-column transfer and Kostka inversion are classical ingredients; their use is distinguished from the new family-specific sign classification. The scope is deliberately fixed: all four attachment groups must be nonempty, and the original graph is not allowed to vary. No general classification of quasi-line graphs is asserted.

1. Introduction

For a finite simple graph G, its chromatic symmetric function is

$$X_G = \sum_{\kappa \text{ proper}} \prod_{v \in V(G)} x_{\kappa(v)}.$$

It is Schur-positive if every coefficient in its Schur expansion is nonnegative. The claw-free Schur-positivity conjecture is false: Prajapati's twelve-vertex line graph has coefficient -64 at $(3,3,3,3)$

[1]. The same pair of order-twelve counterexamples was independently obtained by Matherne and Morales; we retain the chronology and attribution stated in those works. Wang, Zhang and Zhao constructed infinite counterexample families by attaching a clique through one edge [2]. Neither a new base counterexample nor the first infinite counterexample family is claimed here.

Our subject is a different, multiparameter question. Can attachment to several ports repair the negative coefficient, and can one describe the entire Schur-positive region? For the construction below, all coefficient signs are controlled by one quartic. This yields both positive and negative regions and a sharp transition under dilution.

Stable-partition expansions, inclusion–exclusion, and Kostka inversion are established tools. General Schur-coefficient formulas are available [3]; higher independence-number support bounds are also known [4, 5]. Our contribution is the specific complete classification, its finite certificate, and the sharp transition geometry. The general algebraic ingredients are not presented as new in isolation.

2. The graph family and the classification

Let H have the following ordered edges:

$$(0,1),(0,3),(0,4),(0,5),(1,2),(2,3),(2,6),(2,7),(1,8),(3,9),(4,5),(6,7).$$

Let $G = L(H)$, with vertices numbered $0,\dots,11$ in this order, and ports $p_1 = 8, p_2 = 9, p_3 = 10, p_4 = 11$. The ports are independent. Adjoin a clique partitioned into D_1, D_2, D_3, D_4, D_0 of respective sizes a_1, a_2, b_1, b_2, u. Join every vertex of D_i to p_i for $1 \le i \le 4$, and add no other old–new edges. Denote the graph by $\Gamma(a_1, a_2, b_1, b_2; u)$. Throughout the graph-theoretic statements,

$$a_1, a_2, b_1, b_2 \in \mathbb{Z}_{\ge 1}, \qquad u \in \mathbb{Z}_{\ge 0}.$$

Set

$$\alpha = a_1 + a_2, \quad \beta = b_1 + b_2, \quad R = a_1 a_2, \quad S = b_1 b_2, \quad L = \alpha + \beta, \quad t = L + u,$$

and define

$$Q(u) = -u(t-1)(t-2)(t-3) + (R + S + \tfrac{3}{4}\alpha\beta)(t-2)(t-3) - \tfrac{9}{2}(R\beta + S\alpha)(t-3) + 12RS. \quad (2.1)$$

Every neighborhood of Γ is the union of at most two cliques. For a non-port old vertex, use its two endpoint cliques in the line graph; for a port, use its old neighborhood and its attachment group; for a new vertex, use the new clique and its possible port. Thus Γ is quasi-line, and in particular claw-free. Its order is $12 + t$.

Theorem 2.1 (Computer-assisted classification)**.** *Let* $\lambda_* = (4,4,4,4,1^{t-4})$. *Then*

$$[s_{\lambda_*}] X_\Gamma = 64(t-4)!\,Q(u).$$

Every other Schur coefficient is nonnegative. Hence X_Γ *is Schur-positive if and only if* $Q(u) \ge 0$, *and when* $Q(u) < 0$ *the displayed coefficient is its unique negative Schur coefficient. For* $t \ge 12$, *all* 151 *nondistinguished coefficients in the supporting template set defined below are strictly positive.*

The positive-size assumptions are part of the theorem. No classification with a zero attachment group is asserted.

3. The distinguished coefficient

For $J \subseteq [4]$, let G_J be obtained from G by completing the selected ports to a clique, and write $h_J = [s_{(3,3,3,3)}]X_{G_J}$. In exactly t variables, the new clique uses every color once. Inclusion–exclusion over forbidden equalities between a port and a vertex of its group gives

$$X_\Gamma(x_1,\dots,x_t) = e_t \sum_{J\subseteq[4]} (-1)^{|J|}(t-|J|)! \prod_{i\in J} |D_i|\, X_{G_J}(x_1,\dots,x_t). \tag{3.1}$$

At most one vertex in a group can have its port's color. Selected port colors must be distinct, exactly the additional coloring condition of G_J. Once their group vertices are selected, the other clique colors have $(t-|J|)!$ assignments. This proves (3.1).

Multiplication by $e_t = x_1\cdots x_t$ adds one to each of the t padded parts of a Schur index. The resulting map on partitions of length at most t is injective. Therefore

$$[s_{\lambda_*}]X_\Gamma = \sum_J (-1)^{|J|}(t-|J|)! \prod_{i\in J} |D_i|\, h_J. \tag{3.2}$$

This is the multiple-port version of the full-column transfer used in [2, Lemmas 3.2–3.3].

The necessary core counts are as follows. Here N_μ counts unordered stable partitions of type μ.

J	N_{3333}	N_{4332}	N_{4422}	h_J
$\varnothing$ or singleton	32	160	32	−64
$\{1,2\}$ or $\{3,4\}$	32	128	16	64
other pair	32	136	24	48
triple	32	80	0	288
$[4]$	32	0	0	768

For completeness, normalize the colors on vertices $0,1,2,3$ to $0,1,2,3$. The colors on vertices $4,5,6,7$ are permutations c with $c_0 \neq 0, c_1 \neq 1$, so there are 14 possibilities. The allowed port colors are respectively

$$\{0,1,2,3\}\setminus\{0,c_0\},\quad \{0,1,2,3\}\setminus\{1,c_1\},\quad \{0,1\},\quad \{c_0,c_1\}.$$

Require distinct colors on selected ports and count the resulting class sizes. These 224 assignments give the table. Triangular inversion, or the 4×4 Jacobi–Trudi determinant, gives $h_J = 24N_{3333} - 6N_{4332} + 4N_{4422}$. Substitution in (3.2) yields

$$-64t! + 64L(t-1)! + 64(R+S+\tfrac{3}{4}\alpha\beta)(t-2)! - 288(R\beta+S\alpha)(t-3)! + 768RS(t-4)!.$$

Factoring $64(t-4)!$ proves the formula in Theorem 2.1. The auxiliary program `core_jt_check.py` independently reconstructs the sixteen coefficients by enumerating stable partitions and applying Jacobi–Trudi, without the Kostka recurrence used in the full certificate.

4. A finite certificate for all the other coefficients

4.1. Support and transfer

The matching number of H is four: a perfect matching would have to use the two pendant edges and would leave two disjoint odd triangles. The four port edges themselves form a matching. Two old stable sets cover at most eight vertices. Three cover at most ten, since a union of three matchings has degree at most three and must omit an edge at each of the two nonadjacent degree-four vertices

of H. Each stable block contains at most one new clique vertex. Consequently every monomial index obeys

$$\lambda_1 \le 5, \quad \lambda_1+\lambda_2 \le 10, \quad \lambda_1+\lambda_2+\lambda_3 \le 13, \quad \sum_{i=1}^{h} \lambda_i \le 12+h \quad (h \ge 4).$$

For $t \ge 12$ these indices form the dominance ideal

$$\mathcal{T} = \{\nu : |\nu| \le 12,\ \nu_1 \le 4,\ \nu_1+\nu_2+\nu_3 \le 10\},$$

with missing parts interpreted as zero, under the correspondence

$$j_\nu = 12-|\nu|, \quad q_\nu = \ell(\nu), \qquad \lambda_\nu(t) = (\nu_1+1, \dots, \nu_{q_\nu}+1, 1^{t+j_\nu-q_\nu}).$$

The set has 152 members. Kostka unitriangularity preserves this dominance down-set under monomial-to-Schur inversion.

Let $F_{J,\gamma,r}$ count stable partitions of G_J together with a choice of r blocks to augment by one clique vertex, with every selected-port block augmented, such that the nonunit final block sizes are γ. The number N_γ of full stable partitions of that profile is

$$N_\gamma = \sum_{J\subseteq[4]} (-1)^{|J|} \prod_{i\in J} |D_i| \sum_{r\ge|J|} F_{J,\gamma,r}\,(t-|J|)_{\underline{r-|J|}}. \tag{4.1}$$

To see this, restrict a stable partition to the old vertices. The clique vertices assigned to its augmented blocks must be distinct. Inclusion–exclusion forbids assigning a group vertex to its port's block. A set of forbidden assignments requires distinct selected-port blocks, which is imposed by G_J; the remaining injective assignments are counted by the falling factorial. The unused new vertices are singletons. This also proves that the transfer uses no graph-size interpolation.

Write $A_\nu = [s_{\lambda_\nu(t)}]X_\Gamma/(t-12)!$ and $B_\nu = [m_{\lambda_\nu(t)}]X_\Gamma/(t-12)!$. If c_ν is the product of multiplicity factorials of the nonunit parts, then

$$B_\nu = c_\nu N_\nu \prod_{h=1}^{12+j_\nu-q_\nu} (t-12+h). \tag{4.2}$$

At fixed t, each B_ν is multi-affine in the four attachment sizes.

Lemma 4.1 (Long-column Kostka polynomial)**.** *For a dominating template pair $\lambda_\nu(t) \trianglerighteq \lambda_\eta(t)$, the Kostka number $K_{\lambda_\nu(t),\lambda_\eta(t)}$ is a polynomial in $t \ge 12$ of degree at most $\delta = j_\eta - j_\nu \ge 0$.*

Proof. In a semistandard tableau, cells occupied by the first q_η labels form a diagram θ of size $12-j_\eta+q_\eta$ and height at most q_η. All other labels occur once. The number of remaining cells outside the first column is $\delta - q_\eta + \ell(\theta) \le \delta$. Cut the remaining shape after row twelve. Its lower portion is a first-column chain of length $T = t+j_\nu-12 \ge 0$, and the upper poset is fixed. In a linear extension of that upper poset, let h be the number of cells after its last first-column cell, or the whole extension if no first-column cell remains. Those $h \le \delta$ cells are off-column and have no comparisons with the lower chain. The admissible interleavings number $\binom{T+h}{h}$. Sum over upper extensions and the finitely many fillings of θ. □

The polynomial can therefore be computed from $\delta+1$ consecutive exact Kostka values. The certificate also checks one further value. The values are obtained from the usual horizontal-strip recurrence: remove the cells carrying the largest label and recurse. Descending inversion gives

$$A_\eta = B_\eta - \sum_{\nu \trianglerighteq \eta} A_\nu K_{\lambda_\nu(t),\lambda_\eta(t)}. \tag{4.3}$$

It preserves separate multi-affinity in the four sizes at fixed t.

4.2. Reduction to four corners

Automorphisms of the core independently exchange p_1, p_2 and p_3, p_4. Every A_ν, at fixed t, is thus a linear combination of $(1, \alpha, R) \otimes (1, \beta, S)$. In particular it is bilinear in R, S when α, β, t are fixed. Since all four sizes are at least one,

$$\alpha - 1 \le R \le \alpha^2/4, \qquad \beta - 1 \le S \le \beta^2/4.$$

Let $E(\alpha) = (1, \alpha - 1)$ and $B(\alpha) = (\alpha/2, \alpha/2)$. Every coefficient is a convex combination of its evaluations at the four pair endpoints EE, EB, BE, BB. More explicitly, use weights

$$\rho = \frac{R - (\alpha - 1)}{\alpha^2/4 - (\alpha - 1)}, \qquad \sigma = \frac{S - (\beta - 1)}{\beta^2/4 - (\beta - 1)},$$

and coefficients $(1-\rho)(1-\sigma), (1-\rho)\sigma, \rho(1-\sigma), \rho\sigma$. When a denominator vanishes the two endpoints coincide and that weight may be set to zero. Half-integral balanced endpoints are polynomial evaluations, not fractional graphs.

4.3. The finite verification statement

For each endpoint, the following 45 substitutions cover every integer $\alpha, \beta \ge 2$, $u \ge 0$ with $t \ge 12$:

1. $\alpha = 6 + A,\ \beta = 6 + B,\ u = U$;
2. $\alpha = r,\ \beta = 12 - r + A,\ u = U$, for $r = 2, 3, 4, 5$, and the four transposed substitutions;
3. fixed $\alpha, \beta \ge 2$ with $\alpha + \beta \le 11$, and $u = 12 - \alpha - \beta + U$.

All free variables are nonnegative. The first case handles two large pair sums, the second a small pair sum with total at least twelve, and the third the remaining 36 pair sums.

The certificate program `four_port_classification.py` constructs the 16 core tables, the 7,796 relevant Kostka polynomials, and the 152 normalized Schur polynomials by (4.1)–(4.3). It verifies each reconstruction identity and separate multi-affinity. For every nondistinguished coefficient in each context, every nonzero rational coefficient and the constant term are positive. The exact totals are

endpoint	positive terms	least coefficient	least constant
EE	279578	48	9289728000
EB	289224	16	9289728000
BE	289224	16	9289728000
BB	297557	16	13005619200

There are $4 \cdot 45 \cdot 151 = 27,180$ positive polynomial certificates. For $4 \le t < 12$, full stable-partition transfer and Schur inversion over all partitions check the remaining $\sum_{t=4}^{11} \binom{t}{4} = 792$ parameter tuples. In each, no coefficient other than the distinguished one is negative.

These are finite exact statements, regenerated from the displayed graph using integers and rational numbers, with no input tables or private data files. The generator uses the unique block containing the least remaining vertex to enumerate each stable partition once. The stated positivity tests, the proved exhaustive cover, and the four-corner identity prove the remaining assertions of Theorem 2.1. The argument is computer-assisted, not a claimed sign-reversing involution or formal proof-assistant result.

5. A sharp dilution threshold

In this section allow real $a_1, a_2, b_1, b_2 \ge 1$ and $u \ge 0$ in (2.1). Set $E_2 = R+S+3\alpha\beta/4$, $E_3 = R\beta+S\alpha$, $E_4 = RS$.

Lemma 5.1. *$Q(0) \geq 4$ and $Q'(u) < 0$ for $u \geq 0$. Thus Q has a unique positive real zero u_*.*

Proof. Write $x = \alpha - 2$, $y = \beta - 2$ and use the same four corners. Let

$$H_E(z) = \tfrac{z}{2}(5z^2 - 2z + 1), \qquad H_B(z) = \tfrac{z}{2} + \tfrac{z^2}{4} + z^3 + \tfrac{z^4}{4}.$$

At corner ij, $Q(0) = 4 + H_i(x) + H_j(y) + C_{ij}(x, y)$, where the coefficients of C_{ij}, in the order $(xy, x^2y, xy^2, x^3y, xy^3, x^2y^2)$, are

EE	5/2	3/4	3/4	3/4	3/4	3/2
EB	5/2	3/4	9/8	3/4	1/8	5/8
BB	5/2	9/8	9/8	1/8	1/8	1/2

and BE is the transpose of EB. All cross terms are nonnegative, and $5z^2 - 2z + 1 = 5(z - 1/5)^2 + 4/5 > 0$. This proves $Q(0) \geq 4$ by bilinear interpolation.

Expansion gives

$$Q(u) = -u^4 + (6 - 3L)u^3 + c_2u^2 + c_1u + Q(0),$$

where $c_2 = E_2 - 3L^2 + 12L - 11$ and $c_1 = (2L - 5)E_2 - \frac{9}{2}E_3 - (L - 1)(L - 2)(L - 3)$. The bounds $E_2 \leq 5L^2/16$ and $E_3 \geq 3L - 8$ follow from $R \leq \alpha^2/4$, $S \leq \beta^2/4$, $\alpha\beta \leq L^2/4$, and the expansion of the third elementary symmetric function after writing each attachment size as $1 + z_i$. With $v = L - 4 \geq 0$ they give

$$c_2 \leq -43v^2/16 - 19v/2 - 6 < 0, \qquad c_1 \leq -3v^3/8 - v^2/16 - 7v - 9 < 0.$$

Every term of $Q'(u)$ is nonpositive and its constant is negative. Finally $Q(u) \to -\infty$. □

Theorem 5.2 (Sharp normalized threshold)**.** *Let $c = (\sqrt{2}-1)/2$. For every real attachment quadruple in $[1, \infty)^4$, $0 < u_* < cL$. Moreover*

$$\sup_{a_1,a_2,b_1,b_2 \in \mathbb{Z}_{\geq 1}} u_*/L = c,$$

and the supremum is not attained. For integer graph parameters, X_Γ is Schur-positive exactly when $u \leq u_*$.

Proof. Put $r = \sqrt{2}$, $x = \alpha - 2$, $y = \beta - 2$, and

$$d_0 = 18r - 14, \quad d_1 = 53r/2 - 17, \quad d_{11} = 53r/2 - 15,$$

$$\delta = 3 + 27r/8, \quad \epsilon = (51r - 7)/8, \quad k = (3 + 2r)/16, \quad m = (6 + 7r)/16, \quad n = 7r/8.$$

All these constants are positive. Define

$$F_E(z) = (6 + 7r/2)z^2 + 3z^3/8 + kz^4, \qquad F_B(z) = (17r/2 - 5/4)z^2 + (5/8 + 3r/4)z^3.$$

At corner ij, direct expansion of (2.1) gives

$$-Q_{ij}(cL) = d_0 + d_1(x + y) + d_{11}xy + F_i(x) + F_j(y) + T_{ij}(x, y), \tag{5.1}$$

where the coefficients of T_{ij} in the order $(x^2y, xy^2, x^3y, xy^3, x^2y^2)$ are

EE	δ	δ	k	k	0
EB	δ	ϵ	k	m	m
BB	ϵ	ϵ	m	m	n

and BE is the transpose of EB. Thus every corner is strictly negative and bilinear interpolation proves $Q(cL) < 0$. Lemma 5.1 gives the strict upper bound.

For sharpness take $(a_1, a_2, b_1, b_2) = (1, 1, M, M)$, $L = 2M + 2$. For fixed $v \geq 0$,

$$L^{-4}Q(vL) \longrightarrow (1+v)^2(1/4 - v - v^2).$$

The limit is positive for $0 \leq v < c$ and zero at c. For every such v, the unique root eventually exceeds vL. Together with the upper bound, this proves $u_*/L \to c$. The final assertion follows from Theorem 2.1. □

Theorem 5.3 (Asymptotic extremizers)**.** *For any sequence of real attachment quadruples in* $[1, \infty)^4$ *with* $L \to \infty$*, one has* $u_*/L \to c$ *if and only if*

$$\min\{(\alpha + |b_1 - b_2|)/L, (\beta + |a_1 - a_2|)/L\} \longrightarrow 0.$$

Equivalently, the normalized quadruple approaches $\{(0, 0, 1/2, 1/2), (1/2, 1/2, 0, 0)\}$.

Proof. Set $h = \alpha/L$, $g = 1 - h$, $\rho = R/L^2$, $\sigma = S/L^2$. The limiting parameter domain is compact: $0 \leq h \leq 1$, $0 \leq \rho \leq h^2/4$, $0 \leq \sigma \leq g^2/4$. Uniformly on bounded v intervals,

$$L^{-4}Q(vL) = q(v) + O(L^{-1}),$$

where

$$q(v) = -v(1+v)^3 + (\rho + \sigma + 3hg/4)(1+v)^2 - \tfrac{9}{2}(\rho g + \sigma h)(1+v) + 12\rho\sigma.$$

At $v = c$ its four product-corner values, in the order $(0, 0)$, $(0, g^2/4)$, $(h^2/4, 0)$, $(h^2/4, g^2/4)$, are

$$-k(h^3 + g^3), \quad -h(kh^2 + mg^2), \quad -g(kg^2 + mh^2), \quad -hg(m - 3hg/4).$$

For $0 < h < 1$ these are strictly negative, since $hg \leq 1/4$ and $m > 3/16$. At $h = 0$, equality requires $\sigma = 1/4$; at $h = 1$, it requires $\rho = 1/4$. Thus $q(c) = 0$ only at the two stated allocations.

If the normalized roots tend to c, evaluate the uniform limit at each root and pass to any convergent parameter subsequence. Its limit must satisfy $q(c) = 0$. Compactness gives necessity. Conversely, either stated allocation produces the limiting polynomial from the sharpness proof, whose sign is positive below c. Monotonicity and the universal upper bound give sufficiency. Finally $|b_1 - b_2|^2/L^2 = g^2 - 4\sigma$ and the analogous identity for a_1, a_2 give the displayed equivalent condition. □

6. The smallest integral transition

Proposition 6.1 (Computer-assisted minimality)**.** *The least attachment mass L admitting an integral root is* 33*. The least graph order admitting a zero distinguished coefficient is* 48*. At order* 48*, the parameters are precisely the within-pair swaps and pair exchanges of* $(4, 8, 10, 11; 3)$*. These operations are symmetries of Q; no assertion about the number of graph-isomorphism classes is intended.*

Proof. Direct substitution gives

$$Q_{4,8,10,11}(u) = (3 - u)(u^3 + 96u^2 + 2839u + 27050).$$

The cubic is positive on $u \geq 0$, so $u_* = 3$. For minimality enumerate all positive ordered quadruples with $4 \leq L \leq 36$. There are $\binom{36}{4} = 58,905$. For each, integer evaluation of $4Q$ and strict monotonicity isolate the only possible integral zero by binary search between 0 and $\lceil L/2 \rceil$. The endpoint

signs follow from Lemma 5.1 and Theorem 5.2. The exact program `sharp_check.py` finds sixteen ordered zeros, whose representatives under the stated symmetries are exactly $(4, 8, 10, 11; 3)$ and $(2, 14, 3, 15; 4)$. The first has $L = 33$ and order 48; the second has $L = 34$ and order 50. Every smaller possible graph has $L \leq 35$ and is included in the enumeration. Equality at order 48 is covered by $L \leq 36$. □

At the order-48 boundary all other supporting Schur coefficients are positive by Theorem 2.1. The graph is therefore Schur-positive although its distinguished coefficient vanishes. The upper bound $u < cL$ is necessary, but not sufficient, for positivity; the exact test is $Q(u) \geq 0$.

7. Reproducibility and scope

The full classification certificate regenerates its tables from the edge list and uses exact integers and rational numbers. It includes monomial reconstruction identities, all endpoint signs, all 792 small cases, direct stable-partition controls, cell-tableau Kostka controls and hook-length controls. Its finite sign computations are essential to Theorem 2.1. The threshold identities are independent of that large certificate. The separate `sharp_check.py` checks them in $\mathbb{Q}(\sqrt{2})$: each side of (5.1) has separate degree at most four, so equality on a 5×5 rational grid proves the polynomial identity exactly. The four limiting identities similarly have degree at most four in h. No floating-point sign decision is used.

This paper concerns the stated core and strictly positive attachment groups. It does not classify arbitrary quasi-line graphs, extend the classification to zero-sized groups, or claim novelty for stable partition transfer and triangular basis changes. The general Schur-positivity problem is not reduced to a uniformly small computation when the core itself varies.

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